

Ryan Hassell

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PROFESSIONAL EXPERIENCE

Associate Software Engineer – Advanced Artificial Intelligence

June 2025 – Present

Cobbs Creek Healthcare

Newtown Square, PA

- Developed a production-grade AI analytics chatbot supporting natural language querying over 10+ enterprise datasets, generating over 25 visualization types (Sankey, bar, line, heatmap, tables, etc.) with contextual memory.
- Built multi-agent AI workflows (10+ agents) using LangChain/LangFlow for retrieval, SQL planning, visualization selection, and explanation, enabling end-to-end automated analytics pipelines.
- Designed LLM-driven literature review generators and interactive analytics dashboards using TypeScript, R Shiny, and Streamlit, backed by AWS, Azure, Snowflake, and Databricks.
- Collaborated with clinicians and data scientists to integrate agentic AI systems into live research and analytics workflows, improving insight turnaround and usability.
- **Tech Stack:** Python, TypeScript/JavaScript, SQL, LangChain/LangFlow, FastAPI, React, Streamlit, R Shiny, Plotly, AWS, Azure, Snowflake, Databricks.

Software Engineer Intern

June 2023 – June 2024

Insomniac Design

Washington D.C.

- Led backend user and data management architecture for NZDPU platform, enabling secure emissions data exchange for 380+ organizations.
- Engineered RESTful APIs, custom user/file management systems, RBAC, and Firebase Authentication across 197+ jurisdictions; scaled to 1,000+ accounts.
- Optimized SQL queries for 2M+ records, reduced manual workflows by 40%, and ensured compliance across 11 industries using GCP.
- **Tech Stack:** Python, JavaScript, SQL, FastAPI, Firebase, GCP

EDUCATION

Saint Joseph's University

Philadelphia, PA

BS in Computer Science

May 2025

PROJECTS

Virtual Travel Platform

Cobbs Creek Healthcare

Flutter, Python, Firebase, Agora SDK

- Built a real-time global travel app with live guides, using Agora for video and Firebase for messaging and authentication.
- Developed cross-platform client/server and designed upcoming LLM-driven personalization features.

Data Chatbot Interface

Cobbs Creek Healthcare

Python, React, OpenAI API, SQL, LangChain/LangFlow, Plotly

- Developed multi-agent LLM architecture for autonomous SQL query planning, data retrieval, visualization, and explanation.

AI Research Agent & Literature Review Generator

Cobbs Creek Healthcare

Python, LangChain, Streamlit, Snowflake, AWS EC2

- Developed multi-agent LLM platform for end-to-end literature review workflows: screening, extraction, quality assessment, and report generation.

Net-Zero Data Public Utility (NZDPU) Platform

Insomniac Design

Python, GCP, PostgreSQL, FastAPI

- Implemented backend architecture, custom user/file systems, and optimized SQL for 2M+ emissions records, reducing latency by 30% and automating workflows.

LEADERSHIP & ACTIVITIES

Founder & President

2022 – 2025

Computer Science Club, Saint Joseph's University

- Recruited 60+ members and organized workshops, guest lectures, and coding competitions.

TECHNICAL SKILLS

Core Skills: AI/LLM systems, multi-agent workflows, data engineering, full-stack development, cloud analytics.

Languages & Tools: Java, C/C++/C, Python, TypeScript, SQL, JavaScript, AWS/GCP, Snowflake, FastAPI, React, Streamlit, LangChain (additional technologies available upon request).